

RATE ADEQUACY & LOSS DEVELOPMENT ANALYSIS

A Chain-Ladder Study of Five P&C; Insurers' Schedule P Filings, 2016–2025

Companies studied: Allstate • Farmers Insurance • Travelers • Zurich • Liberty Mutual

Lines of business: Combined (All Lines) • Homeowners/Farmowners • Private Passenger Auto Liability/Medical • Commercial Auto/Truck Liability/Medical • Workers' Compensation • Commercial Multiple Peril

Method: cumulative net loss triangles were built from each insurer's Schedule P disclosures, developed to ultimate using volume-weighted loss development factors (the chain-ladder method), and converted into indicated loss ratios by accident year. Each year's indicated loss ratio was measured against that company-line's own 10-year average to flag whether pricing looks overcharged, roughly adequate, or undercharged.

Prepared by Pushkar | Independent actuarial research project | Data source: publicly filed Schedule P annual statement exhibits

Executive Summary

This project turns raw Schedule P loss-triangle data from five major U.S. property-casualty insurers into a repeatable rate-adequacy diagnostic. Across 270 company-line-year observations spanning 2016–2025, the chain-ladder indicated loss ratio was “**roughly adequate**” 54% of the time, **undercharged** 24% of the time, and **overcharged** 21% of the time — but that steady average hides a much more interesting story underneath.

The real finding isn't the individual year-by-year verdicts — it's the **shape of the cycle** they trace out. All five carriers moved through the same three phases at almost the same time, regardless of size or line mix:

- **2020 (pandemic dip):** lockdowns cut claim frequency sharply, industry-average indicated loss ratio fell to 59.4%, and overcharged verdicts spiked to 11 of 27 observations — the highest share of any year except 2025.
- **2022–2023 (inflation shock):** claim severity in auto and commercial lines surged faster than premium could reprice; the industry average jumped to 67.3% and undercharged verdicts hit 14 and 13 observations in those two years — more than half of all lines written.
- **2024–2025 (rate hardening catch-up):** insurers pushed rate increases through faster than loss trends, and by 2025 the market flipped hard — 20 of 27 observations came back overcharged, the most lopsided year in the dataset.

In other words, the data recovers the underwriting cycle directly from public filings, without needing any proprietary pricing data — pandemic under-pricing, inflation-driven under-reserving, and a hard-market correction, all visible in the same ten years.

Headline number

If every company-line combination had priced to its own 10-year average loss ratio every year, none of the 270 observations would show up as more than 5 points off. Instead, 46% of them do — meaning under this chain-ladder view, insurers spend roughly half their time meaningfully away from their own long-run target, not because pricing is bad, but because loss trends move faster than rate filings can follow them.

Data & Methodology

1. Source data. Cumulative net paid/incurred loss triangles for accident years 2016–2025 were pulled from each insurer's Schedule P annual statement exhibits, split into six segments: Combined (All Lines), Homeowners/Farmowners, Private Passenger Auto Liability/Medical, Commercial Auto/Truck Liability/Medical, Workers' Compensation, and Commercial Multiple Peril.

2. Development factors. Volume-weighted age-to-age loss development factors (LDFs) were computed for each triangle and chained into age-to-ultimate factors, the standard chain-ladder approach for estimating how immature losses will grow to their final value.

3. Ultimate losses and loss ratio. Multiplying latest-diagonal losses by the age-to-ultimate factor produced an estimated ultimate loss (and implied reserve) for each accident year; dividing by that year's earned premium gave the indicated loss ratio.

4. Rate adequacy verdict. Each company-line's own 10-year average indicated loss ratio was used as its target/benchmark. A given year is:

- **Roughly Adequate** — within ± 5 points of the average
- **Undercharged** — more than 5 points above the average (losses running hotter than premium supports)
- **Overcharged** — more than 5 points below the average (premium collected exceeds the loss trend)

5. Coverage gaps. Three company-line combinations had insufficient Schedule P detail to support a triangle and were excluded rather than estimated: Allstate Workers' Compensation, Farmers Combined All Lines, and Zurich Private Passenger Auto Liability/Medical. All comparisons below note when a company is missing from a chart for this reason.

Framing note. The benchmark here is each line's own historical average, not a regulator-approved target loss ratio or a competitor's pricing — so “undercharged” and “overcharged” describe deviation from a company's own long-run norm, not an absolute profitability judgment.

Industry-Wide Trends

Averaging the indicated loss ratio across all five companies and all six lines in a given year produces a single industry curve. It is not flat — it traces a clear pandemic-dip, inflation-spike, hard-market-correction arc:

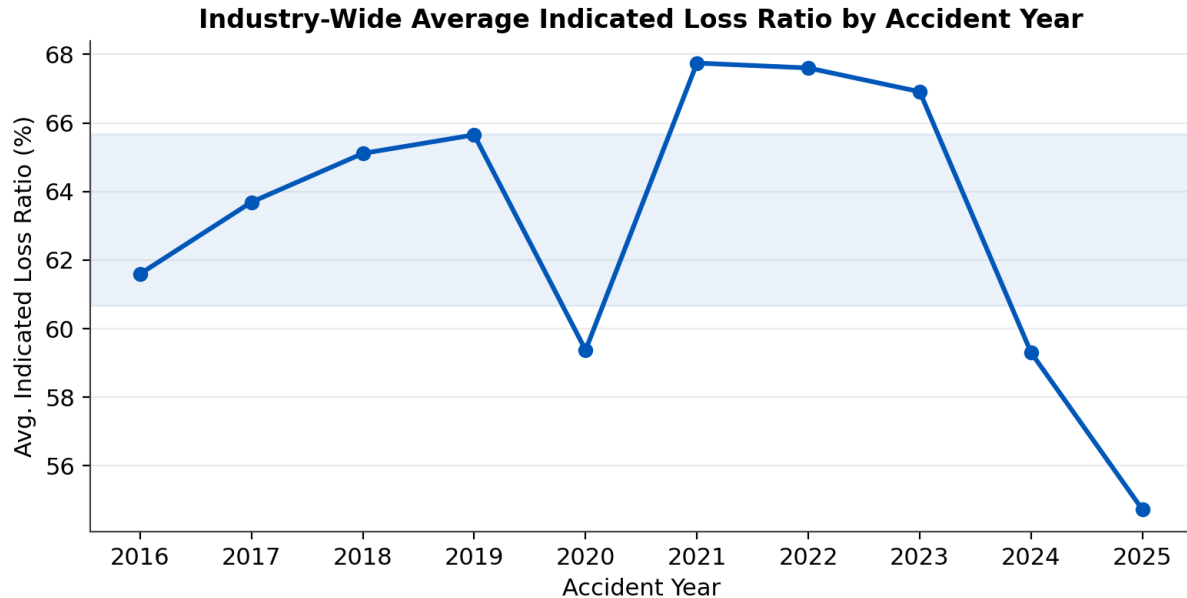


Figure 1. Average indicated loss ratio across all 5 companies × 6 lines, by accident year. Shaded band marks ±2.5 points around the 10-year mean.

The same pattern shows up in verdict counts. Overcharged calls cluster in 2020 and especially 2025; undercharged calls cluster in 2022–2023:

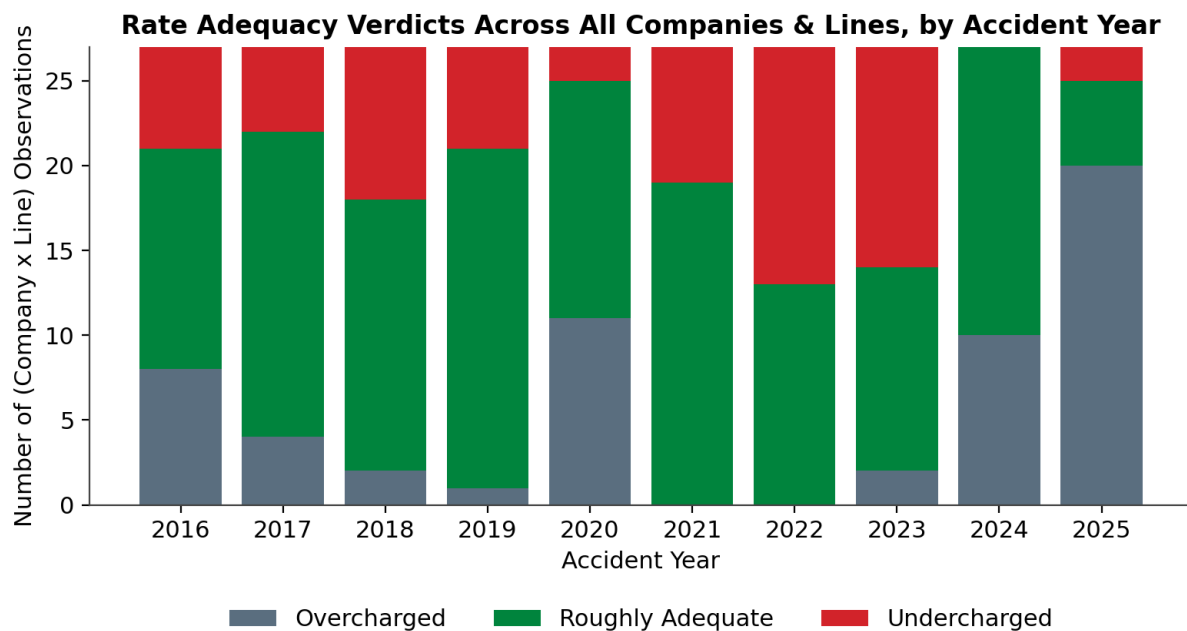


Figure 2. Count of Overcharged / Roughly Adequate / Undercharged verdicts across all company-line observations, by accident year.

Company Comparison

Looking only at each insurer's Combined (All Lines) segment, Zurich consistently runs the highest indicated loss ratio of the group, peaking above 85% in 2019, while Travelers runs the lowest and steadiest, rarely straying far from the high 50s. Farmers has no usable combined-lines triangle in its filings and is omitted from this view.

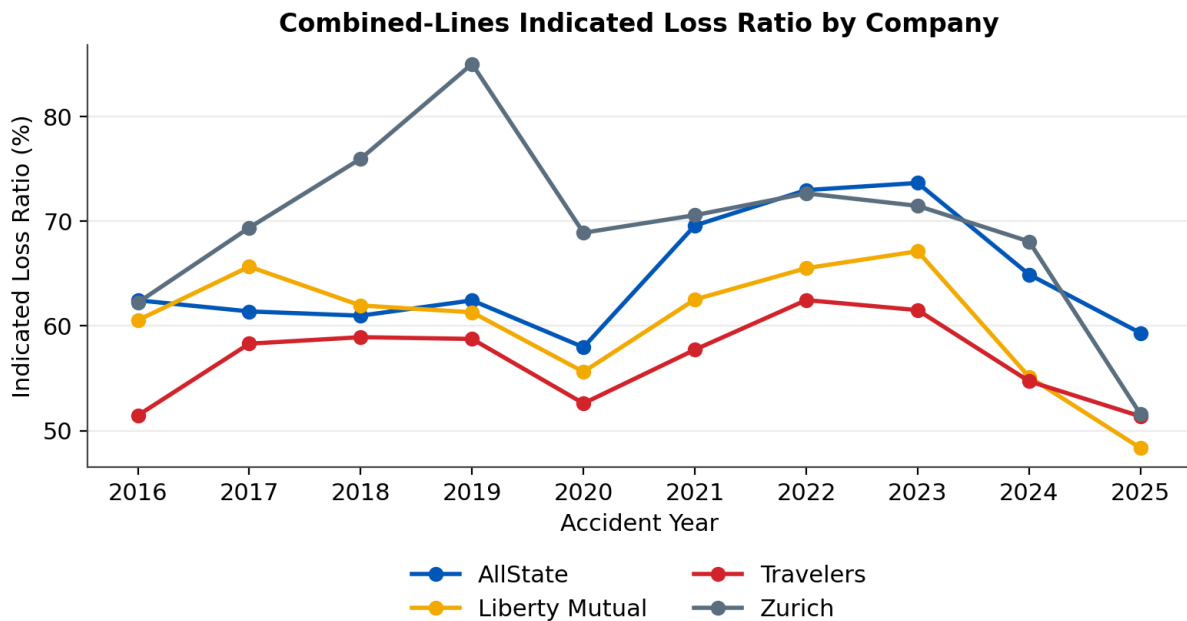


Figure 3. Combined-lines indicated loss ratio by company, 2016–2025 (Farmers excluded — insufficient data).

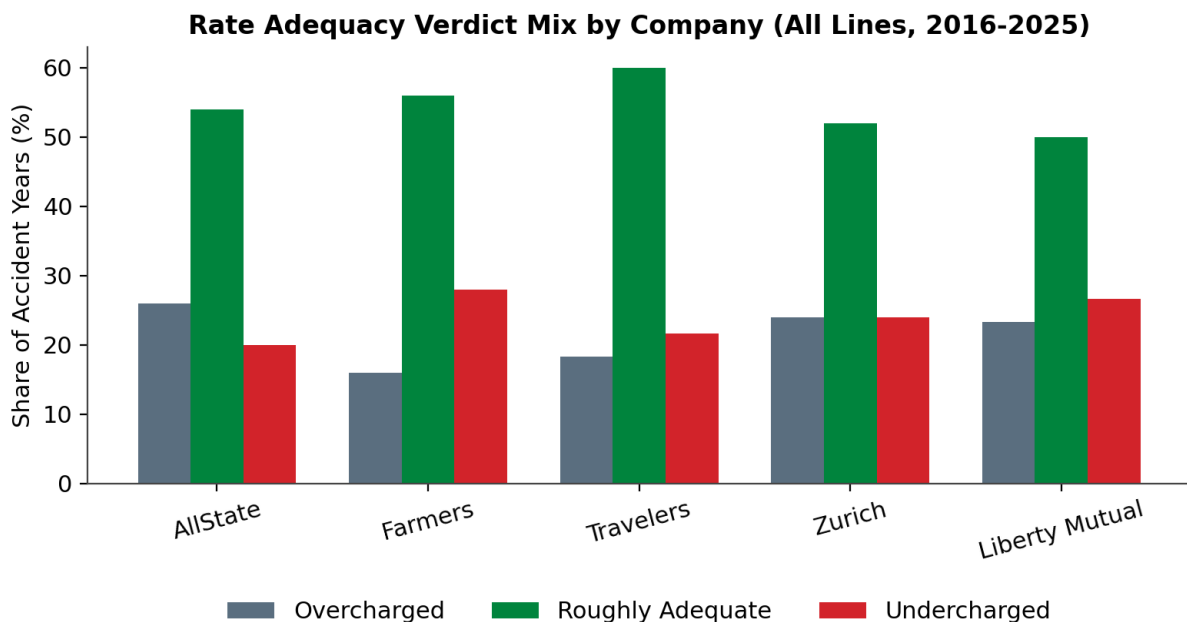


Figure 4. Share of accident-year verdicts by company across all lines. Travelers shows the most “roughly adequate” years (60%); Farmers shows the highest undercharged share (28%).

Verdict Counts by Company (All Lines Combined, 2016–2025)

Company	Overcharged	Roughly Adequate	Undercharged	Total Years Scored
Allstate	13	27	10	50
Farmers	8	28	14	50
Liberty Mutual	14	30	16	60
Travelers	11	36	13	60
Zurich	12	26	12	50

Note: totals differ by company because of the three excluded company-line combinations noted in the methodology (each missing line removes 10 potential observations).

Average Indicated Loss Ratio by Company × Line of Business (%)

Company	Combined	Home/Farm	PP Auto	Comm Auto	Work Comp	Comm Mult. Peri
Allstate	64.6	65.0	73.5	106.3	n/a	54.2
Farmers	n/a	61.7	68.6	55.4	40.5	60.1
Travelers	56.8	58.6	63.3	61.7	46.8	54.4
Zurich	69.6	65.7	n/a	87.6	45.3	74.1
Liberty Mutual	60.4	58.0	63.2	65.9	43.9	57.6

Commercial Auto stands out immediately: Allstate and Zurich both average well above 85% on this line, while every other line for every company sits in the 40s-70s. This is the single clearest cross-company red flag in the dataset (see Notable Findings).

Volatility Analysis

Beyond the direction of the verdict, how *volatile* a company or line is tells its own story. A high standard deviation in indicated loss ratio means the chain-ladder result swings hard from year to year — harder to price confidently, harder to reserve for.

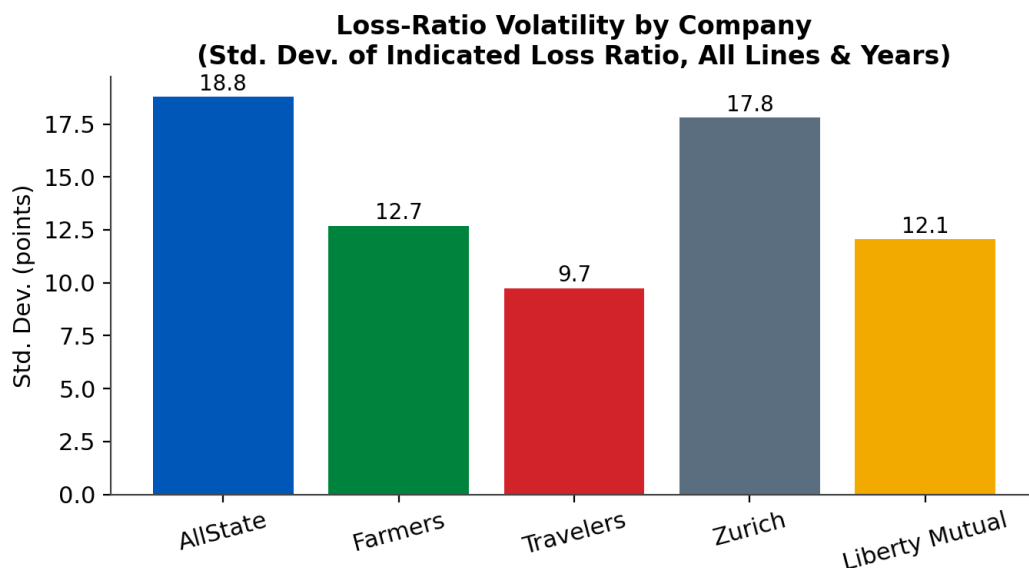


Figure 5. Standard deviation of indicated loss ratio across all lines and years, by company. Allstate and Zurich run nearly twice the volatility of Travelers.

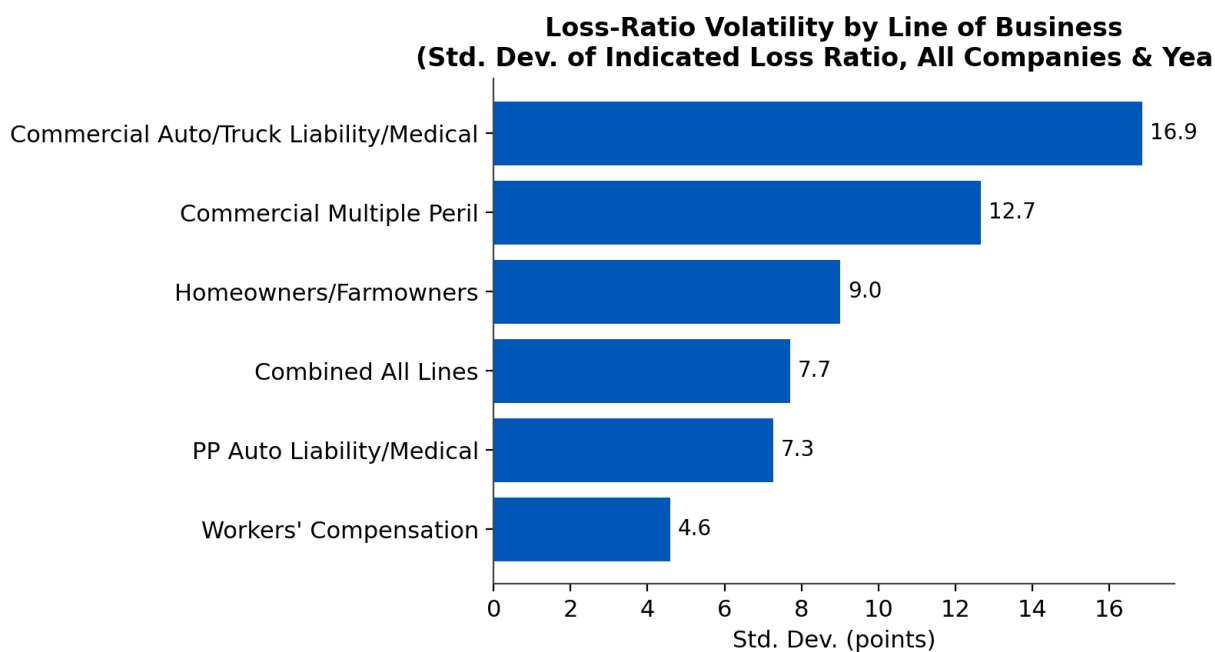


Figure 6. Standard deviation of indicated loss ratio across all companies and years, by line of business. Commercial Auto is by far the most volatile line; Workers' Compensation is the calmest.

Line-of-Business Deep Dive

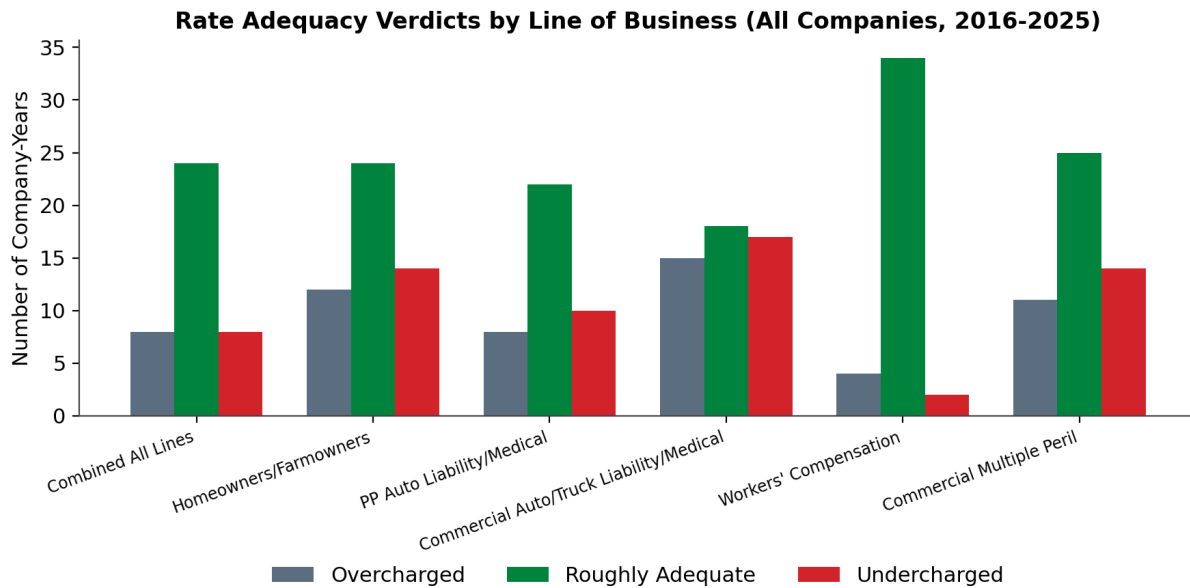


Figure 7. Verdict counts pooled across all five companies, by line of business.

Commercial Auto/Truck Liability/Medical and Commercial Multiple Peril carry the largest share of undercharged verdicts of any line — both are commercial lines exposed to social inflation, nuclear verdicts, and rising vehicle repair/replacement costs, all of which outran premium in 2022–2023. Workers' Compensation, by contrast, is the most stable line in the dataset on every measure.

Homeowners and Private Passenger Auto

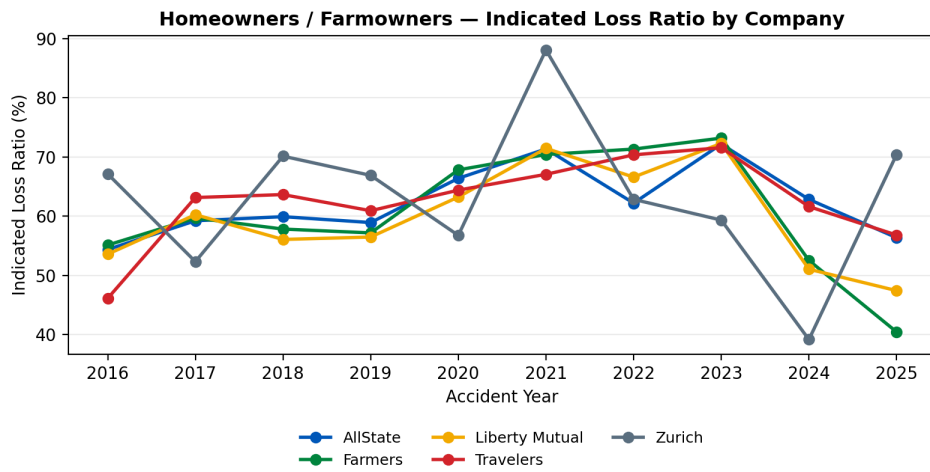


Figure 8. Homeowners/Farmowners indicated loss ratio by company.

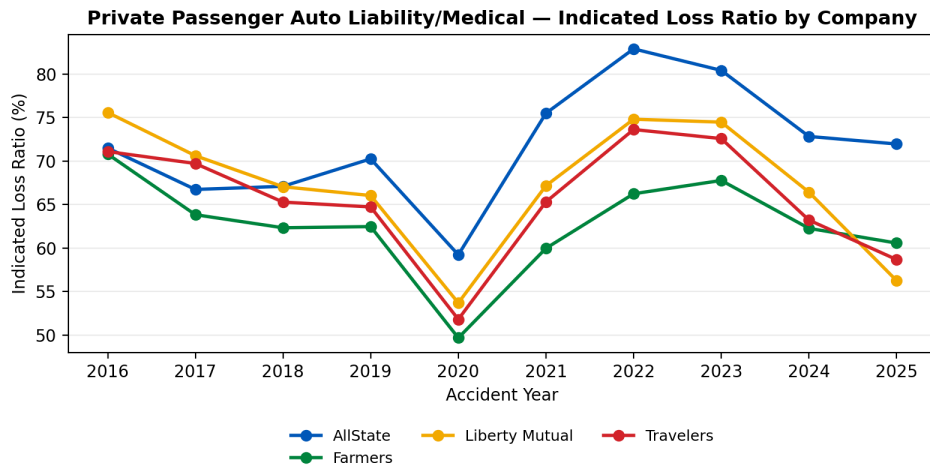


Figure 9. Private Passenger Auto Liability/Medical indicated loss ratio by company (Zurich excluded — insufficient data).

Commercial Auto and Commercial Multiple Peril

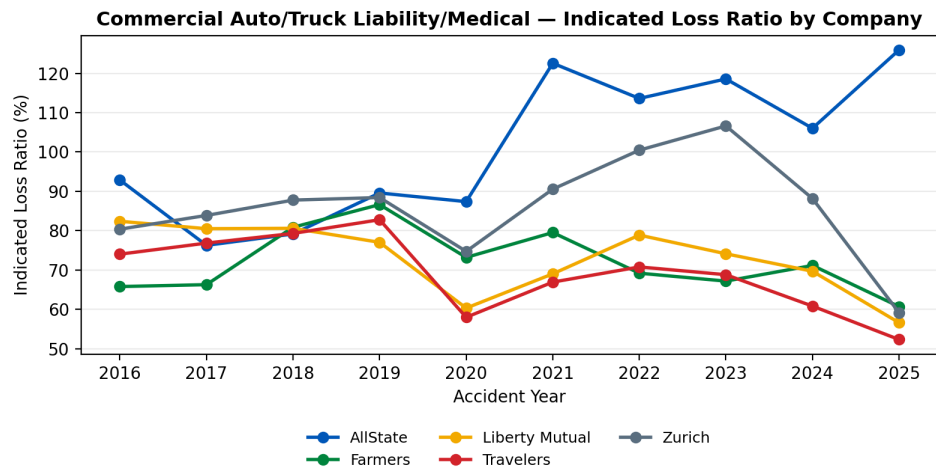


Figure 10. Commercial Auto/Truck Liability/Medical indicated loss ratio by company. Allstate and Zurich both cross 100% in multiple years — this line is written on a small premium base for both companies, which amplifies swings.

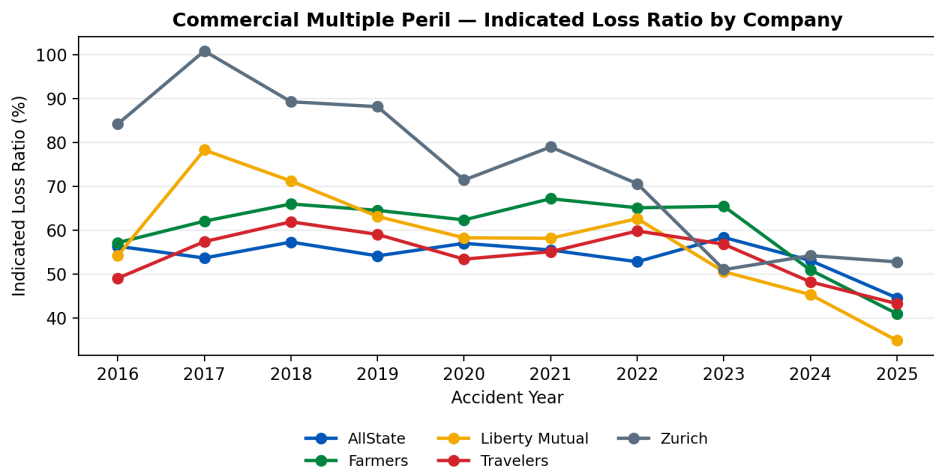


Figure 11. Commercial Multiple Peril indicated loss ratio by company.

Notable Findings

1. Commercial Auto is a red flag for two of five carriers

Allstate and Zurich both average above 85% indicated loss ratio on Commercial Auto/Truck Liability/Medical — roughly 20-40 points above every other line either company writes. Allstate's worst year (2025) hit 125.9% against a 10-year target of 103.8%, and its best year (2017) fell to 76.3% — a 50-point swing on the same line in the same company. This is consistent with a small, thinly-spread commercial book where a handful of large claims move the loss ratio disproportionately.

2. The industry round-trips through the full underwriting cycle in ten years

2020 (pandemic claim-frequency drop) → 2022–2023 (severity/inflation shock, undercharged) → 2025 (hard-market correction, overcharged) is visible in every single company's combined-lines trend, not just the industry average. That consistency across five independent filers, using only public Schedule P data, is itself the most reusable finding of the project — it suggests the method reliably recovers real market cycles rather than company-specific noise.

3. Zurich's single biggest miss was Commercial Multiple Peril in 2017 — 27.4 points undercharged

Zurich's 2017 CMP indicated loss ratio hit 100.8% against a target of 73.4%, the single largest deviation (in either direction) in the entire 270-observation dataset. Its 2021 Homeowners result (88.1% vs. 62.4% target, +25.7 points) is the second largest. Zurich also shows the highest overall volatility of the five companies.

4. Premium growth did not track loss experience 1-for-1

Allstate and Travelers grew combined-lines earned premium 82-87% from 2016 to 2025, while Zurich and Liberty Mutual grew only 34-36% — yet Zurich shows the highest average loss ratio of the group and Liberty Mutual one of the lowest. Slower premium growth did not protect Zurich from running hot, and faster premium growth did not push Allstate or Travelers to systematically overcharge.

5. Workers' Compensation is the outlier — calm, low, and consistently over-reserved for

Across the four companies with usable Workers' Comp data (Farmers, Travelers, Zurich, Liberty Mutual), average indicated loss ratios run 40.5-46.8%, roughly 15-25 points below every other line, with the lowest year-to-year volatility of any segment. This lines up with how WC is typically reserved in practice: long-tail, heavily case-reserved, and priced with wide conservatism margins relative to shorter-tail lines.

6. 2025 is the most one-sided year in the whole dataset

20 of 27 scored observations in 2025 came back "Overcharged" — more than double any other year. If the chain-ladder read is right, this is the clearest single-year signal in the dataset that rate increases pushed through in 2023-2024 may have overshot the loss trend by 2025, at least relative to each line's own 10-year average.

Company Snapshots

Allstate

Highest volatility tied with Zurich (std. dev. 18.8 pts). Commercial Auto is a serious outlier at a 106.3% average loss ratio. Combined lines otherwise runs a fairly typical mid-60s average with a 26%/54%/20% overcharged/adequate/undercharged split.

Farmers

No usable Combined All-Lines triangle in the filings. Highest share of undercharged years of the five (28%), concentrated in Commercial Multiple Peril and Homeowners/Farmowners. Lowest Workers' Comp loss ratio of the group at 40.5%.

Travelers

Lowest volatility of any company (std. dev. 9.7 pts) and the highest share of "roughly adequate" years (60%). Combined lines loss ratio stays in a tight 51-62% band across all ten years — the most consistently priced book in the dataset.

Zurich

Highest average loss ratio and second-highest volatility of the five. Holds both the single largest undercharge (2017 CMP, +27.4 pts) and one of the largest overcharges (2025 Commercial Auto, -26.8 pts) in the dataset — a company that swings hard in both directions. No usable PP Auto triangle.

Liberty Mutual

Second-lowest volatility (std. dev. 12.1 pts). Slowest premium growth of the group (35.9% over the decade) paired with a below-average combined loss ratio, suggesting a comparatively conservative, slower-growing book relative to peers.

Limitations & Caveats

- **Own-average benchmark, not a true rate-need estimate.** "Undercharged"/"overcharged" here means deviation from a company-line's own 10-year mean, not a formal actuarial indicated rate change, which would also need trend, expense, and profit-load assumptions.
- **Volume-weighted LDFs only.** No tail factor, Bornhuetter-Ferguson blending, or credibility weighting was layered on top of the pure chain-ladder result; small-premium lines (e.g., Allstate's Commercial Auto book) are especially sensitive to this simplification.
- **Three company-lines excluded outright** (Allstate Workers' Comp, Farmers Combined, Zurich PP Auto) rather than estimated, per the underlying workbooks — comparisons involving those companies are based on the remaining five lines only.
- **Net losses, not gross.** Reinsurance cessions affect net loss triangles differently across companies depending on each insurer's reinsurance program, which is not visible in Schedule P net exhibits alone.
- **Ten accident years is a short window** for a line like Workers' Compensation or Commercial Auto, where true tail development can run well beyond 10 years; the earliest accident years (2016-2018) are the most mature and most reliable, the most recent years (2023-2025) the least.

Conclusions & Where This Could Go Next

This project demonstrates that a chain-ladder rate-adequacy screen, built entirely from public Schedule P filings, is sensitive enough to recover a real, shared underwriting cycle across five independent insurers — a pandemic-driven dip, an inflation-driven spike, and a hard-market correction, all timed consistently across companies. That is a meaningful result on its own: it shows the method is picking up genuine market signal, not company-specific noise.

It also surfaces two clear, line-specific stories worth a closer look: Commercial Auto's outsized loss ratios at Allstate and Zurich, and Workers' Compensation's unusual stability across every company that reports it.

Natural next steps to strengthen this into a portfolio-ready piece:

- Layer in a Bornhuetter-Ferguson estimate alongside the pure chain ladder, especially for the most recent, least-mature accident years, and show how much the verdicts change.
- Add a expected-vs-actual industry benchmark (e.g., A.M. Best or NAIC countrywide loss ratios by line) instead of each company's own average, to distinguish company-specific mispricing from industry-wide cycle effects.
- Extend the excluded company-lines (Allstate WC, Farmers Combined, Zurich PP Auto) using a supplementary data source if available, to complete the 5×6 matrix.
- Automate the workbook-to-verdict pipeline in Python (pandas + a simple chain-ladder function) so new accident years or new companies can be added without rebuilding formulas by hand in Excel.
- Turn the six "Notable Findings" into the spine of a short write-up for a portfolio, cover letter, or interview conversation — each one is a self-contained, defensible story with a number attached.

End of report.